Matrices, Determinants and Inverses

A gentle start, from zero

Teaching notes for an introductory machine-learning class

The aim

By the end of these notes you will be able to read a matrix, multiply two matrices, compute a determinant, and find the inverse of a 2×2 or 3×3 matrix by hand. You need nothing beyond GCSE algebra: multiplying, adding and solving simple equations. Every idea comes with a worked example you can check with a pencil.

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1 What is a matrix?

A **matrix** is a rectangular table of numbers, written inside square brackets:

$$A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \end{bmatrix}.$$

That is all it is. Matrices are useful because they let us store lots of numbers neatly and do the same calculation on all of them at once. In machine learning, a dataset is usually stored as a matrix: one row per example (a person, a house, an image) and one column per feature (age, floor area, pixel brightness).

1.1 Size: rows first, then columns

The **size** of a matrix is written rows × columns, always **rows first**. The matrix A above has 2 rows and 3 columns, so it is a 2×3 matrix (say "two by three").

A memory aid: rows go *across* like rows of seats in a cinema; columns go *up and down* like the columns holding up a building.

Worked example

$$B = \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 5 & 6 \end{bmatrix} \text{ is } 3 \times 2, \qquad C = \begin{bmatrix} 9 & 1 & 0 & 4 \end{bmatrix} \text{ is } 1 \times 4, \qquad D = \begin{bmatrix} 2 \\ 7 \end{bmatrix} \text{ is } 2 \times 1.$$

1.2 Entries

The number in row i and column j is called the (i, j) **entry**, written a_{ij} (lower-case letter, row number first, then column number).

Worked example

In $A = \left[\begin{smallmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \end{smallmatrix}\right]$:

- $a_{11} = 1$ (row 1, column 1);
- $a_{13} = 7$ (row 1, column 3);
- $a_{22} = 5$ (row 2, column 2);
- a_{31} does not exist, because A has no third row.

1.3 Some special matrices

Name	What it is	Example
Square	same number of rows as columns	$\left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$
Vector	a matrix with just one column (or one row)	$\left[\begin{smallmatrix} 2 \\ 7 \end{smallmatrix}\right]$
Zero matrix 0	every entry is 0	$\left[\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right]$
Identity matrix I	square, 1s on the main diagonal, 0s elsewhere	$I_2 = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$, $I_3 = \left[\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right]$

The **main diagonal** of a square matrix runs from the top-left corner to the bottom-right corner. The identity matrix I is the matrix version of the number 1: we will see that multiplying by it changes nothing.

1.4 The transpose

The **transpose** of A, written A^{T}, swaps rows and columns: row 1 becomes column 1, row 2 becomes column 2, and so on. A 2×3 matrix becomes 3×2.

$$A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \end{bmatrix} \qquad \Longrightarrow \qquad A^{\mathsf{T}} = \begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 7 & 8 \end{bmatrix}.$$

2 Adding and scaling matrices

These two operations work exactly as you would guess: entry by entry.

Adding. Two matrices of the *same size* are added by adding matching entries. Matrices of different sizes cannot be added.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 0 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1+5 & 2+0 \\ 3-1 & 4+2 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 2 & 6 \end{bmatrix}.$$

Subtraction works the same way.

Multiplying by a number (called a **scalar**). Multiply every entry:

$$3 \begin{bmatrix} 1 & -2 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 3 & -6 \\ 0 & 12 \end{bmatrix}.$$

Worked example

Let $A = \left[\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right]$ and $B = \left[\begin{smallmatrix} 1 & 1 \\ 4 & -1 \end{smallmatrix}\right]$. Find $2A - B$.
First $2A = \left[\begin{smallmatrix} 4 & 2 \\ 0 & 6 \end{smallmatrix}\right]$. Then

$$2A - B = \begin{bmatrix} 4-1 & 2-1 \\ 0-4 & 6-(-1) \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ -4 & 7 \end{bmatrix}.$$

3 Matrix multiplication

Multiplying two matrices is **not** done entry by entry. It uses a "row times column" recipe. It looks strange at first, but it is exactly the calculation that comes up again and again in practice.

3.1 A motivating example

A café sells coffee at £2.50 and cake at £3.00. On Monday it sells 3 coffees and 2 cakes; on Tuesday, 5 coffees and 1 cake. What is the takings each day?

- Monday: $3 \times 2.50 + 2 \times 3.00 = 13.50$.

- Tuesday: $5 \times 2.50 + 1 \times 3.00 = 15.50$.

Each answer is "multiply matching numbers and add up". Matrix multiplication is exactly this, organised:

$$\underbrace{\begin{bmatrix} 3 & 2 \\ 5 & 1 \end{bmatrix}}_{\text{quantities (row = day)}} \quad \underbrace{\begin{bmatrix} 2.50 \\ 3.00 \end{bmatrix}}_{\text{prices}} = \begin{bmatrix} 3 \times 2.50 + 2 \times 3.00 \\ 5 \times 2.50 + 1 \times 3.00 \end{bmatrix} = \begin{bmatrix} 13.50 \\ 15.50 \end{bmatrix}.$$

3.2 Step 1: check the sizes (the domino rule)

You can only multiply A by B (in that order) if the number of **columns of** A equals the number of **rows of** B. The answer's size comes from the two outside numbers:

$$\underbrace{(m \times \boxed{n})}_{A} \; \underbrace{(\boxed{n} \times s)}_{B} = \underbrace{(m \times s)}_{AB}.$$

Think of dominoes: the touching ends must match.

Worked examples

Sizes	Allowed?	Size of answer
$(2 \times 2)(2 \times 1)$	yes: $2 = 2$	2×1
$(2 \times 3)(3 \times 2)$	yes: $3 = 3$	2×2
$(3 \times 2)(2 \times 3)$	yes: $2 = 2$	3×3
$(1 \times 3)(3 \times 1)$	yes: $3 = 3$	1×1 (a single number)
$(2 \times 3)(2 \times 3)$	**no**: $3 \neq 2$	—

3.3 Step 2: row times column

To find the entry in row i, column j of AB:

1. take **row** i **of** A and **column** j **of** B;
2. multiply them together pair by pair (first with first, second with second, ...);
3. add up the results.

Worked example: a 2×2 times a 2×2

Find AB where $A = \left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$ and $B = \left[\begin{smallmatrix} 5 & 6 \\ 7 & 8 \end{smallmatrix}\right]$. Sizes: $(2 \times 2)(2 \times 2)$, so the answer is 2×2 and has four entries to find.
Top-left (row 1 of A, column 1 of B):

$$\begin{bmatrix} \mathbf{1} & \mathbf{2} \\ 3 & 4 \end{bmatrix} \begin{bmatrix} \mathbf{5} & 6 \\ \mathbf{7} & 8 \end{bmatrix}: \qquad 1 \times 5 + 2 \times 7 = 5 + 14 = 19.$$

Top-right (row 1 of A, column 2 of B): $1 \times 6 + 2 \times 8 = 6 + 16 = 22$.
Bottom-left (row 2 of A, column 1 of B):

$$\begin{bmatrix} 1 & 2 \\ \mathbf{3} & \mathbf{4} \end{bmatrix} \begin{bmatrix} \mathbf{5} & 6 \\ \mathbf{7} & 8 \end{bmatrix}: \qquad 3 \times 5 + 4 \times 7 = 15 + 28 = 43.$$

Bottom-right (row 2 of A, column 2 of B): $3 \times 6 + 4 \times 8 = 18 + 32 = 50$.
So

$$AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}.$$

Worked example: a 2×3 times a 3×2

Find AB where

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 3 & 1 \\ 2 & 1 \\ 1 & 0 \end{bmatrix}.$$

Sizes: $(2 \times 3)(3 \times 2)$; the inner 3s match, so the answer is 2×2. Each entry now adds up three products.

- Row 1, column 1: $1 \times 3 + 0 \times 2 + 2 \times 1 = 3 + 0 + 2 = 5$.
- Row 1, column 2: $1 \times 1 + 0 \times 1 + 2 \times 0 = 1$.
- Row 2, column 1: $(-1) \times 3 + 3 \times 2 + 1 \times 1 = -3 + 6 + 1 = 4$.
- Row 2, column 2: $(-1) \times 1 + 3 \times 1 + 1 \times 0 = 2$.

$$AB = \begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix}.$$

Worked example: a matrix times a vector

$$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \times 4 + 1 \times 5 \\ 1 \times 4 + 3 \times 5 \end{bmatrix} = \begin{bmatrix} 13 \\ 19 \end{bmatrix}.$$

A 2×2 matrix times a 2×1 vector gives a 2×1 vector. In machine learning this is how a model makes predictions for every example at once.

3.4 Order matters!

With ordinary numbers, $3 \times 5 = 5 \times 3$. With matrices, AB **and** BA **are usually different**. Sometimes one of them does not even exist.

Worked example

With $A = \left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$ and $B = \left[\begin{smallmatrix} 5 & 6 \\ 7 & 8 \end{smallmatrix}\right]$ from before, we found $AB = \left[\begin{smallmatrix} 19 & 22 \\ 43 & 50 \end{smallmatrix}\right]$. Now the other way round:

$$BA = \begin{bmatrix} 5 \times 1 + 6 \times 3 & 5 \times 2 + 6 \times 4 \\ 7 \times 1 + 8 \times 3 & 7 \times 2 + 8 \times 4 \end{bmatrix} = \begin{bmatrix} 23 & 34 \\ 31 & 46 \end{bmatrix} \neq AB.$$

So always say which side you multiply on: "multiply B on the left by A" means AB.

3.5 The identity matrix changes nothing

For any matrix A of the right size, $AI = A$ and $IA = A$, just like $1 \times 5 = 5$.

Worked example

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} 1\cdot 3 + 0\cdot(-2) & 1\cdot 8 + 0\cdot 5 \\ 0\cdot 3 + 1\cdot(-2) & 0\cdot 8 + 1\cdot 5 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ -2 & 5 \end{bmatrix}.$$

3.6 Rules that still work, and one that does not

These familiar rules **still hold** (as long as the sizes fit):

- Brackets can be regrouped: $(AB)C = A(BC)$.
- Brackets can be expanded: $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$.
- Numbers can move around: $A(3B) = 3(AB)$.

Common mistake

$AB = BA$ is usually **false** (see above). A second surprise: $AB = 0$ does *not* mean $A = 0$ or $B = 0$.

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1-1 & -1+1 \\ 1-1 & -1+1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Neither matrix is zero, but their product is. This is one reason we cannot simply "divide" by a matrix.

4 Determinants

The **determinant** is a single number calculated from a *square* matrix. It tells us one crucial thing: whether the matrix has an inverse. If the determinant is 0, it does not. We write it $\det(A)$ or $|A|$.

4.1 The 2×2 determinant

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

In words: **multiply down the main diagonal, then subtract the product of the other diagonal**.

Worked examples

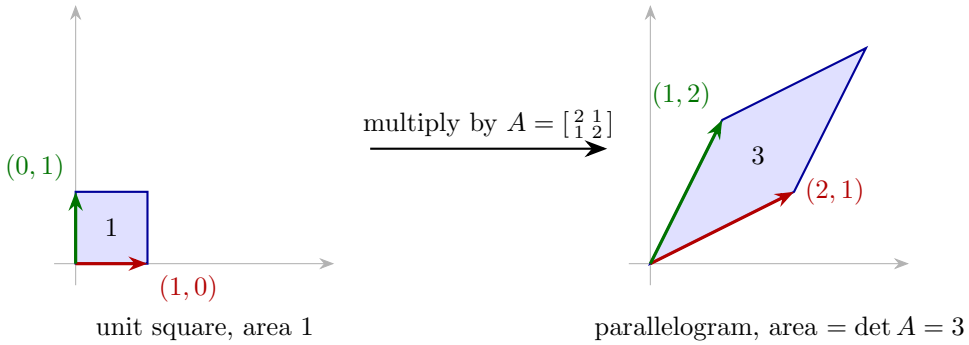
- $\det\left[\begin{smallmatrix} 3 & 1 \\ 4 & 2 \end{smallmatrix}\right] = 3 \times 2 - 1 \times 4 = 6 - 4 = 2.$
- $\det\left[\begin{smallmatrix} 2 & 5 \\ 1 & 3 \end{smallmatrix}\right] = 2 \times 3 - 5 \times 1 = 6 - 5 = 1.$
- $\det\left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right] = 1 \times 4 - 2 \times 3 = 4 - 6 = -2.$ (Negative answers are fine.)
- $\det\left[\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right] = 1 \times 4 - 2 \times 2 = 4 - 4 = 0.$

4.2 What does the determinant mean?

Think of a 2×2 matrix as a machine that moves every point of the plane: the point (x, y) is sent to $A\left[\begin{smallmatrix} x \\ y \end{smallmatrix}\right]$. The two columns of A tell you where the points $(1, 0)$ and $(0, 1)$ land, so the unit square (area 1) is turned into a parallelogram built on the two columns.

Key idea

The determinant is the **area scale factor**: every shape's area is multiplied by $|\det A|$. A negative determinant means the shape is also flipped over (like a mirror image).



unit square, area 1 parallelogram, area $= \det A = 3$

Check: $\det\left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right] = 2 \times 2 - 1 \times 1 = 3$. The unit square's area 1 becomes area 3.

Worked examples of the meaning

- $\left[\begin{smallmatrix} 2 & 0 \\ 0 & 3 \end{smallmatrix}\right]$ stretches across by 2 and up by 3, so the unit square becomes a 2×3 rectangle of area 6. And indeed $\det = 2 \times 3 - 0 \times 0 = 6$.
- $\left[\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right]$ sends $(1, 0)$ to $(1, 2)$ and $(0, 1)$ to $(2, 4)$. These point in the *same direction*, so the "parallelogram" is squashed flat onto a line: area 0. And $\det = 0$.

Why $\det = 0$ means trouble. A matrix with determinant 0 squashes the whole plane onto a line (or a point). Many different points land in the same place, so there is no way to work backwards and recover where a point came from. That is exactly why such a matrix has **no inverse** (Section 5).

4.3 The 3×3 determinant

For a 3×3 matrix we break the problem into three 2×2 determinants. Two new words first.

Minor. The minor M_{ij} is the 2×2 determinant you get by **covering up row i and column j**.

Worked example: finding a minor

For $B = \left[\begin{smallmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{smallmatrix}\right]$, find M_{11}. Cover row 1 and column 1:

$$\begin{bmatrix} \mathbf{1} & \mathbf{2} & \mathbf{3} \\ \mathbf{0} & 1 & 4 \\ \mathbf{5} & 6 & 0 \end{bmatrix} \quad \Longrightarrow \quad M_{11} = \det \begin{bmatrix} 1 & 4 \\ 6 & 0 \end{bmatrix} = 1 \times 0 - 4 \times 6 = -24.$$

Cofactor. The cofactor C_{ij} is the minor with a sign attached, following a chessboard pattern that starts with $+$ in the top-left corner:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix} \qquad \text{so} \qquad C_{ij} = \pm M_{ij}.$$

So $C_{11} = +M_{11}$, $C_{12} = -M_{12}$, $C_{13} = +M_{13}$, and so on.

The rule. Pick any row (the first row is the usual choice). Multiply each entry by its cofactor and add:

$$\det A = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}.$$

Written out in full:

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}.$$

Worked example: a 3×3 determinant

Find $\det B$ for $B = \left[\begin{smallmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{smallmatrix}\right]$, expanding along row 1.

1. Entry 1: cover row 1, column 1. Minor $= \det\left[\begin{smallmatrix} 1 & 4 \\ 6 & 0 \end{smallmatrix}\right] = 0 - 24 = -24$. Sign $+$. Contribution: $+1 \times (-24) = -24$.
2. Entry 2: cover row 1, column 2. Minor $= \det\left[\begin{smallmatrix} 0 & 4 \\ 5 & 0 \end{smallmatrix}\right] = 0 - 20 = -20$. Sign $-$. Contribution: $-2 \times (-20) = +40$.
3. Entry 3: cover row 1, column 3. Minor $= \det\left[\begin{smallmatrix} 0 & 1 \\ 5 & 6 \end{smallmatrix}\right] = 0 - 5 = -5$. Sign $+$. Contribution: $+3 \times (-5) = -15$.

Add up: $\det B = -24 + 40 - 15 = 1$.

Shortcut: choose a row (or column) with zeros. You may expand along *any* row or column, using the same chessboard of signs, and you always get the same answer. Zeros make terms vanish.

Worked example: using a zero-heavy row

Find $\det A$ for $A = \left[\begin{smallmatrix} 2 & 1 & 1 \\ 1 & 3 & 2 \\ 1 & 0 & 0 \end{smallmatrix}\right]$. Row 3 is $(1, 0, 0)$, so expand along row 3 (signs $+$, $-$, $+$):

$$\det A = +1 \times \det \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix} - 0 \times (\ldots) + 0 \times (\ldots) = 1 \times (2 - 3) = -1.$$

One small determinant instead of three.

The meaning carries over: a 3×3 determinant is the **volume** scale factor, and $\det = 0$ means 3D space is squashed flat onto a plane, a line or a point.

4.4 Useful facts about determinants

Fact	Why it helps
$\det I = 1$	the identity changes nothing, so it scales area by 1
$\det(AB) = \det A \times \det B$	doing two moves multiplies their scale factors
$\det(A^{\mathsf{T}}) = \det A$	you can expand along columns as well as rows
a row or column of zeros $\Rightarrow \det = 0$	expand along that row: every term is 0
two equal (or proportional) rows $\Rightarrow \det = 0$	the shape is squashed flat
$\det(kA) = k^n \det A$ for an $n \times n$ matrix	every one of the n rows is multiplied by k

Worked example: checking $\det(AB) = \det A \times \det B$

With $A = \left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$ and $B = \left[\begin{smallmatrix} 5 & 6 \\ 7 & 8 \end{smallmatrix}\right]$: $\det A = 4 - 6 = -2$ and $\det B = 40 - 42 = -2$. We found $AB = \left[\begin{smallmatrix} 19 & 22 \\ 43 & 50 \end{smallmatrix}\right]$, and $\det(AB) = 19 \times 50 - 22 \times 43 = 950 - 946 = 4$. Indeed $(-2) \times (-2) = 4$.

5 The inverse of a matrix

5.1 What is an inverse?

For numbers, the inverse of 5 is $\frac{1}{5}$, because $5 \times \frac{1}{5} = 1$. It *undoes* multiplication by 5. The number 0 has no inverse: nothing times 0 gives 1.

For matrices, the role of 1 is played by the identity I. The **inverse** of a square matrix A is the matrix A^{-1} that undoes A:

$$AA^{-1} = I \qquad \text{and} \qquad A^{-1}A = I.$$

When does an inverse exist?

A square matrix has an inverse **if and only if** $\det A \neq 0$.

- $\det A \neq 0$: A is called **invertible** (or non-singular).
- $\det A = 0$: A is called **singular**. It squashes space flat, so it cannot be undone.

Only **square** matrices can have an inverse.

Notation warning. A^{-1} does *not* mean "1 divided by every entry". It is a whole new matrix, found by the methods below.

5.2 The 2×2 inverse: a formula

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

A recipe to remember it:

1. Find the determinant $ad - bc$. If it is 0, stop: there is no inverse.
2. **Swap** the two entries on the main diagonal (a and d).
3. **Change the sign** of the other two entries (b and c).
4. **Divide** every entry by the determinant.

Worked example: a 2×2 inverse

Find the inverse of $A = \left[\begin{smallmatrix} 4 & 7 \\ 2 & 6 \end{smallmatrix}\right]$.

1. Determinant: $4 \times 6 - 7 \times 2 = 24 - 14 = 10$. Not zero, so the inverse exists.
2. Swap 4 and 6: $\left[\begin{smallmatrix} 6 & 7 \\ 2 & 4 \end{smallmatrix}\right]$.
3. Change the signs of 7 and 2: $\left[\begin{smallmatrix} 6 & -7 \\ -2 & 4 \end{smallmatrix}\right]$.
4. Divide by 10:

$$A^{-1} = \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}.$$

Always check by multiplying back:

$$AA^{-1} = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix} = \begin{bmatrix} 2.4-1.4 & -2.8+2.8 \\ 1.2-1.2 & -1.4+2.4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \checkmark$$

Why does the formula work? Multiply A by the swapped-and-signed matrix:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} ad-bc & -ab+ba \\ cd-dc & -cb+da \end{bmatrix} = \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix} = (ad-bc)\, I.$$

So dividing by $ad-bc$ gives exactly I. It also shows why $ad-bc=0$ is fatal: we would have to divide by 0.

Worked example: no inverse

$A = \left[\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right]$ has $\det A = 4-4 = 0$. It is singular: no inverse exists. (We saw in Section 4.2 that it squashes the plane onto a line.)

5.3 The 3×3 inverse, method 1: cofactors

This method reuses the minors and cofactors from Section 4.3. The recipe has four steps:

1. **Determinant.** Find $\det A$. If it is 0, stop.
2. **Matrix of cofactors.** Replace every entry by its cofactor C_{ij} (cover its row and column, take the 2×2 determinant, apply the chessboard sign).
3. **Adjugate.** Transpose the matrix of cofactors (swap rows and columns). This is called the **adjugate**, $\operatorname{adj} A$.
4. **Divide** by the determinant:
$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A.$$

(The 2×2 formula is exactly this recipe applied to a 2×2 matrix.)

Worked example: a 3×3 inverse by cofactors

Find the inverse of

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 2 \\ 1 & 0 & 0 \end{bmatrix}.$$

Step 1: determinant. We found $\det A = -1$ in Section 4.3. Not zero, so the inverse exists.
Step 2: all nine minors (cover row i and column j, take the 2×2 determinant):

$$M_{11} = \det\left[\begin{smallmatrix} 3 & 2 \\ 0 & 0 \end{smallmatrix}\right] = 0 \qquad M_{12} = \det\left[\begin{smallmatrix} 1 & 2 \\ 1 & 0 \end{smallmatrix}\right] = -2 \qquad M_{13} = \det\left[\begin{smallmatrix} 1 & 3 \\ 1 & 0 \end{smallmatrix}\right] = -3$$

$$M_{21} = \det\left[\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right] = 0 \qquad M_{22} = \det\left[\begin{smallmatrix} 2 & 1 \\ 1 & 0 \end{smallmatrix}\right] = -1 \qquad M_{23} = \det\left[\begin{smallmatrix} 2 & 1 \\ 1 & 0 \end{smallmatrix}\right] = -1$$

$$M_{31} = \det\left[\begin{smallmatrix} 1 & 1 \\ 3 & 2 \end{smallmatrix}\right] = -1 \qquad M_{32} = \det\left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right] = 3 \qquad M_{33} = \det\left[\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right] = 5$$

Now apply the chessboard of signs $\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$ to get the cofactors:

$$\text{cofactors} = \begin{bmatrix} +0 & -(-2) & +(-3) \\ -0 & +(-1) & -(-1) \\ +(-1) & -(3) & +(5) \end{bmatrix} = \begin{bmatrix} 0 & 2 & -3 \\ 0 & -1 & 1 \\ -1 & -3 & 5 \end{bmatrix}.$$

Quick check: row 1 of A times row 1 of the cofactors gives the determinant: $2 \times 0 + 1 \times 2 + 1 \times (-3) = -1$. ✓

Step 3: adjugate (transpose: row 1 becomes column 1, and so on):

$$\operatorname{adj} A = \begin{bmatrix} 0 & 0 & -1 \\ 2 & -1 & -3 \\ -3 & 1 & 5 \end{bmatrix}.$$

Step 4: divide by $\det A = -1$ (that is, change every sign):

$$A^{-1} = \frac{1}{-1} \begin{bmatrix} 0 & 0 & -1 \\ 2 & -1 & -3 \\ -3 & 1 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ -2 & 1 & 3 \\ 3 & -1 & -5 \end{bmatrix}.$$

Check $AA^{-1} = I$, row by column:

- Row 1 $(2, 1, 1)$: with column 1, $0 - 2 + 3 = 1$; column 2, $0 + 1 - 1 = 0$; column 3, $2 + 3 - 5 = 0$.
- Row 2 $(1, 3, 2)$: with column 1, $0 - 6 + 6 = 0$; column 2, $0 + 3 - 2 = 1$; column 3, $1 + 9 - 10 = 0$.
- Row 3 $(1, 0, 0)$: picks out row 1 of A^{-1}, which is $(0, 0, 1)$.

The result is I. ✓

5.4 The 3×3 inverse, method 2: row reduction

This method works for any size of matrix and is how computers do it. Write A and I side by side, then use **row operations** to turn the left half into I. Whatever the right half has become is A^{-1}:

$$\left[\, A \,\middle|\, I \,\right] \quad \longrightarrow \quad \left[\, I \,\middle|\, A^{-1} \,\right].$$

You are allowed exactly three kinds of row operation, and you must apply each one to the *whole* row (both halves):

1. **Swap** two rows.
2. **Multiply** a row by a non-zero number.
3. **Add** a multiple of one row to another row.

The usual plan: get a 1 in the top-left corner, use it to make zeros below it; then a 1 in the middle, with zeros above and below; then the same for the last column.

Worked example: the same matrix by row reduction

Start with $A = \left[\begin{smallmatrix} 2 & 1 & 1 \\ 1 & 3 & 2 \\ 1 & 0 & 0 \end{smallmatrix}\right]$ next to I. (R_1, R_2, R_3 name the rows.)

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 3 & 2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \end{array}\right]$$

Swap R_1 and R_3 to get a 1 in the top-left corner:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 3 & 2 & 0 & 1 & 0 \\ 2 & 1 & 1 & 1 & 0 & 0 \end{array}\right]$$

Clear the first column: $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - 2R_1$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 3 & 2 & 0 & 1 & -1 \\ 0 & 1 & 1 & 1 & 0 & -2 \end{array}\right]$$

Swap R_2 and R_3 to get a 1 in the middle:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & -2 \\ 0 & 3 & 2 & 0 & 1 & -1 \end{array}\right]$$

Clear below it: $R_3 \to R_3 - 3R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & -2 \\ 0 & 0 & -1 & -3 & 1 & 5 \end{array}\right]$$

Make the last pivot 1: $R_3 \to -R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & -2 \\ 0 & 0 & 1 & 3 & -1 & -5 \end{array}\right]$$

Clear above it: $R_2 \to R_2 - R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 & 1 & 3 \\ 0 & 0 & 1 & 3 & -1 & -5 \end{array}\right]$$

The left half is now I, so the right half is the inverse:

$$A^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ -2 & 1 & 3 \\ 3 & -1 & -5 \end{bmatrix},$$

exactly what the cofactor method gave.

What if there is no inverse? With row reduction you find out along the way: at some point a whole row of the left half becomes zeros, and you can never turn it into a row of I. That happens exactly when $\det A = 0$.

5.5 Useful facts about inverses

Fact	In words
$(A^{-1})^{-1} = A$	undoing the undo gets you back to A
$(AB)^{-1} = B^{-1}A^{-1}$	the order **reverses**: socks on then shoes, but shoes off then socks
$\det(A^{-1}) = \dfrac{1}{\det A}$	if A scales area by 5, its inverse scales area by $\frac{1}{5}$
$I^{-1} = I$	"do nothing" is undone by "do nothing"

5.6 Why we care: solving equations

Simultaneous equations can be written as one matrix equation $A\mathbf{x} = \mathbf{b}$. If A has an inverse, multiply both sides on the left by A^{-1}:

$$A^{-1}A\mathbf{x} = A^{-1}\mathbf{b} \quad \Longrightarrow \quad I\mathbf{x} = A^{-1}\mathbf{b} \quad \Longrightarrow \quad \mathbf{x} = A^{-1}\mathbf{b}.$$

Worked example: simultaneous equations

Solve $2x + y = 5$ and $x + y = 3$.
In matrix form:

$$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}.$$

The determinant is $2 - 1 = 1$, so the inverse is $\left[\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right]$. Then

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 - 3 \\ -5 + 6 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Check: $2(2) + 1 = 5$ and $2 + 1 = 3$. ✓

If $\det A = 0$, the equations have either no solution or infinitely many, never exactly one. For example, $x + 2y = 3$ and $2x + 4y = 6$ are really the same equation twice, and $\det\left[\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right] = 0$.

This is exactly how linear regression is solved in machine learning: the "best fit" coefficients come from solving a matrix equation of this form.

6 Practice questions

Try it yourself

1. Which of these products exist, and what size is the answer? (a) $(2 \times 3)(3 \times 4)$ (b) $(3 \times 2)(3 \times 2)$ (c) $(1 \times 2)(2 \times 1)$
2. Compute $\left[\begin{smallmatrix} 2 & -1 \\ 0 & 3 \end{smallmatrix}\right]\left[\begin{smallmatrix} 1 & 4 \\ 2 & -2 \end{smallmatrix}\right]$.
3. Find the determinant and the inverse of $\left[\begin{smallmatrix} 5 & 2 \\ 3 & 1 \end{smallmatrix}\right]$. Check your answer.
4. For which value of k does $\left[\begin{smallmatrix} k & 2 \\ 3 & 6 \end{smallmatrix}\right]$ have no inverse?

5. Find $\det\begin{bmatrix} 2 & 0 & 1 \\ 1 & 3 & 2 \\ 1 & 1 & 1 \end{bmatrix}$. Does this matrix have an inverse?
6. Use an inverse matrix to solve $3x + 2y = 7$ and $x + y = 3$.

Answers

1. (a) Yes, 2×4. (b) No: $2 \neq 3$. (c) Yes, 1×1 (a single number).
2. Row 1: $2 - 2 = 0$ and $8 + 2 = 10$. Row 2: $0 + 6 = 6$ and $0 - 6 = -6$. Answer: $\begin{bmatrix} 0 & 10 \\ 6 & -6 \end{bmatrix}$.
3. $\det = 5 - 6 = -1$. Swap, change signs, divide by -1: $\frac{1}{-1}\begin{bmatrix} 1 & -2 \\ -3 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 3 & -5 \end{bmatrix}$. Check: $\begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} -1 & 2 \\ 3 & -5 \end{bmatrix} = \begin{bmatrix} -5+6 & 10-10 \\ -3+3 & 6-5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
4. $\det = 6k - 6$, which is 0 when $k = 1$.
5. Expand along row 1: $2(3 - 2) - 0 + 1(1 - 3) = 2 - 2 = 0$. No inverse. (Notice that row 1 + row 2 $= (3, 3, 3)$, which is 3 times row 3, so the rows are linked and space is squashed flat.)
6. $\det\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} = 3 - 2 = 1$, inverse $\begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$. Then $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} 7 \\ 3 \end{bmatrix} = \begin{bmatrix} 7-6 \\ -7+9 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. So $x = 1$, $y = 2$.

7 One-page summary

Everything on one page

Size. rows × columns. a_{ij} is the entry in row i, column j.
Adding and scaling. Entry by entry. Same size needed to add.
Multiplying AB. Columns of A must equal rows of B; $(m \times n)(n \times s) = m \times s$. Entry (i, j) = row i of A times column j of B, multiplied pair by pair and added. $AB \neq BA$ in general. $AI = IA = A$.
2×2 **determinant.**

$$\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

It is the area scale factor. $\det = 0$ means space is squashed flat.
3×3 **determinant.** Expand along a row: entry × minor, with signs $\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$. Pick the row with the most zeros.
Inverse. $AA^{-1} = A^{-1}A = I$. Exists only for square matrices with $\det A \neq 0$.
2×2 **inverse.** Swap the diagonal, change the signs of the other two, divide by the determinant:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

3×3 **inverse.** Either: determinant → cofactors → transpose (adjugate) → divide by the determinant. Or: row-reduce $[\,A \mid I\,]$ to $[\,I \mid A^{-1}\,]$.
Always check by multiplying: AA^{-1} should be I.
Solving $A\mathbf{x} = \mathbf{b}$: $\mathbf{x} = A^{-1}\mathbf{b}$.